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Comparison of biogas recovery from MSW using different aerobic-anaerobic operation modes

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ABSTRACT

Aeration pretreatment was demonstrated as an efficient technology to promote methane recovery from a bioreactor landfill with high food waste content. In this study, a short-term experiment was conducted to investigate the effects of aerobic-anaerobic operation modes on biogas recovery. Three landfill-simulated columns (anaerobic control (A1), a constant aeration (C1) and a gradually reduced aeration (C2)) were constructed and operated for 130 days. The aeration frequency was adjusted by oxygen consumption in an aerated MSW landfill. After aerobic pretreatment was halted, the methanogenic phase was rapidly developed in both the C1 and C2 columns, reducing the volatile fatty acid (VFA) concentrations and increasing pH. The methane volumes per dry MSW produced from the C1 and C2 columns were approximately 62 L/kg $_{VS}$ and 75 L/kg $_{VS}$, respectively, while methane produced from the A1 column was almost negligible. The result clearly showed that aerobic pretreatment with gradual reduction of aeration rates could not only improve methane recovery from waste decomposition, but also enhance leachate COD and VFA removal.

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1. Introduction

The bioreactor landfill has gained significant attention in recent years due to increased municipal solid waste (MSW) generation. The main advantages of bioreactor landfills over conventional landfills include a relatively short period of waste stabilization, high biogas production rates, and recovery of landfill space (Ahmadifar et al., 2015; Elagroudy et al., 2008; Mertoglu et al., 2006; Reinhart et al., 2002; Shao et al., 2008; Warith, 2002). However, despite the numerous advantages of bioreactor landfills, high food waste content in the MSW stream in developing countries may make it difficult to construct and operate bioreactor landfills that utilize traditional methods. Elevated concentrations of volatile fatty acids (VFAs) as a result of leachate recirculation may cause retardation of growth methanogenic bacteria, subsequently reducing methane production.

It has been reported that controlled aeration pretreatment could be a useful means to reduce the impact of VFAs on maintaining the balance between hydrolytic and methanogenic reactions (Hua et al., 2005). Favorable conditions for methane generation can be created through aeration by increasing leachate pH and decomposing organic acids generated from the waste hydrolysis process. Xu et al. (2014) demonstrated that the methanogenic phase was quickly established in aerobically pretreated bioreactors, while methane generation was inhibited in an anaerobic bioreactor. In addition to rapidly restoring the methanogenic phase, aerobic pretreatment was also applied to enhance biogas recovery. However, determining aeration rates or frequencies for pretreating MSW is complicated because the performance of aeration relies on the amount of aeration, air distribution in MSW (air permeability and moisture content), air-liquid contact area and reaeration, microbial growth conditions, biodegradable organic content of MSW, etc. In addition, due to the heterogeneous property of MSW and the dynamic changes of MSW with decomposition, it is difficult to formulate the aeration pretreatment parameters (aeration rates, modes, and frequencies) with MSW properties and/or leachate quality (Cossu et al., 2015; Nag et al., 2015; Xu et al., 2015).

Numerous studies have been conducted using various aeration positions, times and frequencies to improve the efficiency of biogas recovery (Botheju and Bakke, 2011; Sang et al., 2009; Wu et al., 2014). Gerassimidou et al. (2013) compared the total methane production from MSW (inoculated with aerobic sludge) by applying different durations of aeration pretreatment (0, 8, 45, and 90 days). The authors reported that the aerobically treated MSW reached the steady methanogenic phase faster than raw MSW. However,

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excessive aerobic pretreatment might lead to the loss of organic matter and consequently the total volume of methane production could be reduced.

Therefore, it is critical to optimize the aeration pretreatment conditions to maximize biogas recovery from MSW. Xu et al. (2015) conducted a study to investigate the effects of two constant aeration frequencies on methane generation in simulated hybrid bioreactors. They argued that the use of high frequency aeration in a simulated bioreactor stimulated methane generation but reduced total methane production due to the loss of organic matter. Based on previous research, it was hypothesized that a gradual reduction in the frequency of aeration pretreatment could be helpful with respect to minimizing the loss of organic matter and increasing the efficiency of aeration pretreatment. Xu et al. (2015) determined the duration of aeration based on leachate pH by aerating MSW until the pH became 7. One of the drawbacks of the pH-based aeration controlling method was that leachate quality (pH) might not properly represent the physicochemical conditions of MSW decomposition. In this study, the aeration pretreatment conditions were determined by oxygen consumption from MSW decomposition; methane generations from MSW decomposition under these conditions were compared focusing on the effects of a constant aeration frequency and a gradual reduction in aeration frequency on landfill performance. To achieve this goal, a short-term experiment was conducted using three laboratory-scale lysimeters.

2. Materials and method

2.1. Experiment setup

Waste components were collected from the Shenzhen University Town (Shenzhen, China) and synthesized to represent the typical MSW composition in China. The compositions of MSW are shown in Table 1. Each component of MSW was shredded to less than 2 cm and mixed well before it was loaded into each reactor. The initial moisture content of the synthesized waste was 44%; a total of 800 mL of deionized water was added to generate initial leachate for recirculation.

Three reactors (A1, C1, C2) were constructed to simulate bioreactors under different operational conditions in one anaerobic and two aerobically pretreated bioreactors. All columns were made of transparent polyacrylic material with a diameter of 10 cm and height of 60 cm. A schematic diagram of those bioreactors is shown in Fig. 1. Each bioreactor was loaded with 2 kg of mixed MSW with a thickness of 40 cm. The final bulk density of the compacted waste was 640 kg/m^3. A 5-cm thick gravel layer was placed on top of each bioreactor for even distribution of recirculated leachate. Another gravel layer was placed at the bottom of the column for leachate drainage. For the aerobically pretreated bioreactors (C1 and C2), a 5-cm thick gravel layer was placed as a separated aeration layer at the 2nd/3rd height of MSW layer (Fig. 1b). Air was introduced to the aeration layer through the 0.5-cm diameter tube connected to the aeration layer. Two ports were installed on top of each bioreactor for leachate recirculation and gas collection. An additional port was installed at the bottom of the column for leachate collection.

In the aerobically pretreated bioreactors (C1 and C2), air was intermittently introduced into each reactor by a compressor (HAILEA, China, 25 W) connected to the aeration pipes, with a flow rate of 150 mL/min (134 mL/kg dry mass/min) for the first 50 days. During each aeration, air was injected for two hours. The aeration frequency in C1 was maintained at a constant frequency of 4 times/d, while C2 was aerated in a gradually reduced mode based on the change of oxygen concentration. All simulated bioreactor landfills were covered by insulation material and operated for 130 days; the temperature was maintained at 30 ± 2 °C using heating tapes with temperature controllers. Leachate collected from each column was recirculated with a flow rate of 150 mL/d. The aeration was stopped at day 50; at this point reactors C1 and C2 were operated under anaerobic conditions. Leachate recirculation was suspended when aeration was ceased in order to avoid the potential inhibition of methanogenic bacterial growth caused by abrupt changes in landfill conditions. Leachate was recirculated again at day 84, when the leachate pH in the pretreated bioreactors was stabilized at approximately 7.

In this research, several different levels of aeration rates (frequencies) were applied to the C2 column to explore the effects of variable aeration rates changed by oxygen demands on biogas recovery. A total of three different levels of aeration frequencies were applied two, three and four times a day. The oxygen demands were estimated by oxygen depletion for 30 min after aeration was halted, assuming that the oxygen levels (% in volume) in landfill gas would be reduced by waste decomposition. The change in oxygen levels within 30 min (ΔO_2/30 min) was calculated using the O_2 concentration between two points: right after aeration was stopped and 30 min after stopping aeration. ΔO_2/30 min was used to determine the time to change the aeration frequency; if ΔO_2/30 min was 2% less than in the previous observations, the aeration frequencies were reduced to the lower level. Detailed operations of each bioreactor landfill are shown in Table 2.

2.2. Sampling and analysis

The gas and leachate of each reactor were sampled regularly. The gas samples were analyzed for O_2 and CH_4 using a gas chromatograph (FULI GC 9790, Hangzhou, China) with a current of 110 mA. The temperatures of the thermal conductivity detector (TCD), sample injection port and capillary column were set at 100 °C, 70 °C and 50 °C, respectively. Leachate samples were analyzed for pH (Sartorius, PB-10, Germany). COD was measured according to the Environmental Protection Industry Standard of the People's Republic of China (HJ/T 399-2007). VFA was determined by spectrophotometry at the wavelength of 500 nm, according to routine analysis of biogas fermentation. A more detailed method description can be found elsewhere (Siedlecka et al., 2008). Blanks and calibration check samples were performed as appropriate.

Table 1
Component of MSW used in simulated bioreactors.

Waste component	Weight percent (%)	Moisture content (%)	Wet mass (kg)	TS (kg)	VS/TS[a]
Food waste	55.0	76	1.10	0.26	0.94
Paper	10.0	8	0.20	0.18	0.86
Plastics	10.0	<1	0.20	0.20	0.92
Sand	20.0	3	0.40	0.38	0.03
Glass	4.50	<1	0.09	0.09	<0.01
Metal	0.50	<1	0.01	0.01	0.08
Total	100	44	2.00	1.12	0.70

[a] The ratio of volatile solid to total solid.

3. Results and discussion

3.1. Controlling aerobic pretreatment based on oxygen consumption rate

The changes in ΔO_2/30 min over time are presented in Fig. 2. In general, the changes in oxygen levels in landfill gas within 30 min decreased over time from 6% to below 3%, indicating that a lower amount of oxygen is required for MSW decomposition. In other

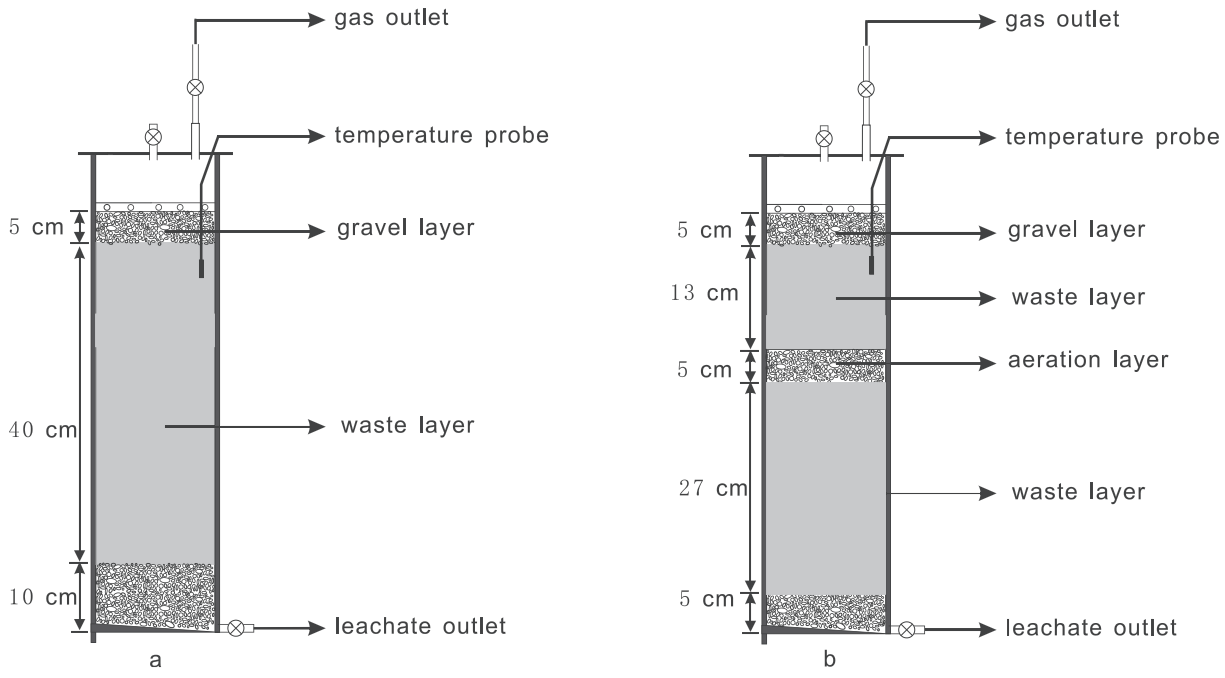


Fig. 1. Schematic design of the simulated columns: (a) anaerobic bioreactor and (b) aerobically pretreated bioreactor.

Table 2
Operations of each simulated bioreactor.

Bioreactor	Aeration frequency		Leachate recirculation (mL/d)		
	0–50 d	51–130 d	0–50 d	51–83 d	84–130 d
A1	No	No	150	No	150
C1	4 times/d	No	150	No	150
C2	• 0–10 d, 4 times/d • 11–17 d, 3 times/d • 18–50 d, 2 times/d	No	150	No	150

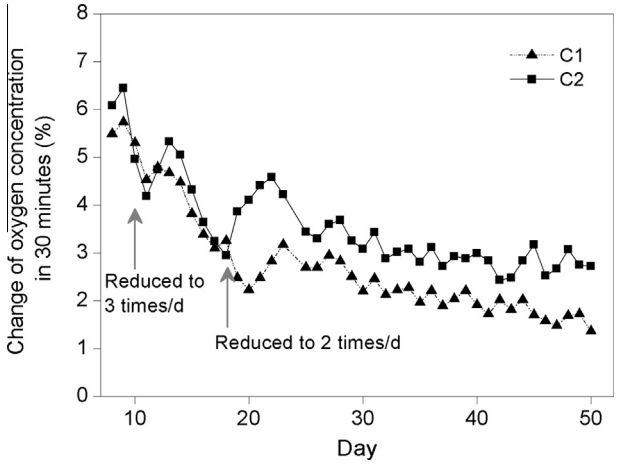


Fig. 2. Changes in the first 30-min oxygen consumption after aeration was stopped.

words, the efficiency of aerobic activity was reduced during the aerobic pretreatment period (Lee et al., 2015; Maria et al., 2014). In the C2 column, as shown in Fig. 2, the aeration frequency was reduced to 3 times/d and 2 times/d at day 11 and day 17, respectively. When the aeration frequency was reduced, the 30-min oxygen consumption increased to some extent for several days, but decreased over time. The ΔO_2/30 min of the C2 was stabilized at approximately 3% after day 30. However, the ΔO_2/30 min of the C1 column continued to decrease to below 2%. Considering the total amount of oxygen consumed within the unit time period, the C2 column showed relatively high oxygen consumption efficiency. These results suggest that changing aeration rates by waste decomposition stages would be required not only to maintain the high efficiency of aerobic decomposition, but also to avoid excessive aeration that may cause the waste to be dried through loss of moisture through air flow.

3.2. Leachate quality of bioreactors

Fig. 3 presents the change of leachate pH, COD and VFAs over time. At the beginning of the experiment, immediate pH drops were observed in all bioreactors (Fig. 3(a)). With the high content of easily biodegradable organic matter (55% food waste), the formation of organic acids could further lower the pH of leachate. The pH of the leachate of all the reactors was approximately 3.7–3.8 on day 9. Such a low pH as a result of the hydrolysis of readily-decomposable organic matter has been frequently reported in other studies (Charles et al., 2009; Sang et al., 2009; Valencia et al., 2009; Warith, 2002). After the initial pH drop, the pH slowly increased over time but still remained in acidic conditions (pH around 5.0) in anaerobic bioreactor A1 during the experiment. By contrast, aeration pretreatment improved the buffer capacity of leachate and reduced the acidic phase of the biodegradation process (Bilgili et al., 2007; Erses and Onay, 2008; Sang et al., 2009). The leachate pH in the C1 and C2 columns quickly increased to over pH 5.0 during the intermittent aeration period. At the end of this experiment, pH values in both the C1 and C2 columns were above 7.

As shown in Fig. 3(b) and (c), COD and VFA concentrations quickly built up in all bioreactors in the first 10 days, reaching around 50,000 mg/L and 25,000 mg/L, respectively, because the

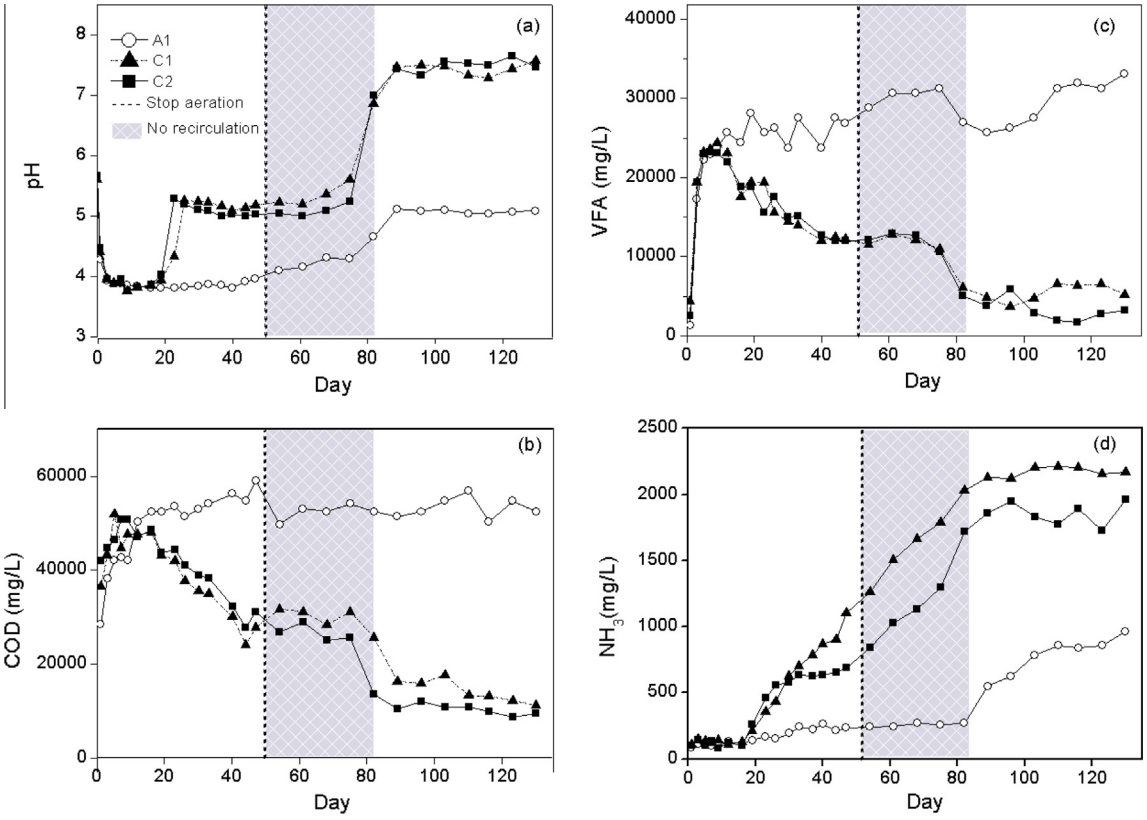


Fig. 3. The change of leachate quality in simulated bioreactors: (a) pH, (b) COD, (c) VFA and (d) NH_3.

rate of formation of hydrolytic products in leachate exceeded the rate of hydrolytic products consumption in the beginning of the experiment (Liu et al., 2014; Sandip et al., 2012; Tatsi and Zouboulis, 2002). The accumulation of hydrolytic products could inhibit further microbial hydrolysis, delaying the consumption of hydrolysis products under anaerobic conditions (Aguilar et al., 1995; Suchowska-Kisielewicz et al., 2012). As a result, the COD and VFA concentrations of A1 maintained over 52,000 mg/L and 33,000 mg/L until the end of this experiment. However, the aerobic pretreatment (aeration for 50 days) resulted in about 50% of COD reduction and 55% of VFA reduction in both the C1 and C2 columns. During the methane generation phase, the concentrations of COD and VFAs in C1 and C2 continued to decrease and settled at around 10,000 mg/L and 4000 mg/L at the end of this experiment, respectively.

The variations of leachate ammonia-nitrogen concentrations are presented in Fig. 3(d). As shown in Fig. 3(d), the ammonia-nitrogen concentration in the anaerobic bioreactor slowly accumulated and reached about 1000 mg/L at the end of the experiment. Due to the aeration pretreatment, the ammonia nitrogen in both C1 and C2 rapidly increased and reached approximately 2000 mg/L at the end of the experiment. The changing profile of ammonia-nitrogen concentrations in this study is comparable to a previous study reported by Xu et al. (2015). In addition, the ammonia nitrogen concentrations in C1 were generally higher than those in C2, which was in accordance with the previous study (Xu et al., 2015). This indicates that aeration may enhance biodegradation of ammonia-containing waste, such as protein, resulting in increasing ammonia–nitrogen concentrations in leachate. However, the oxygen supplied in each column might not be sufficient to remove ammonia though the nitrification-denitrification process, as shown in the study by Xu et al. (2015).

3.3. Methane generation

Fig. 4 presents the methane concentrations, methane generation rates and cumulative methane volume. No methane was generated in the anaerobic bioreactor (A1). The results suggested that A1 underwent severe inhibition of methanogens due to harsh acidic conditions (Fig. 4(a)). This phenomenon was often observed in the degradation of wet MSW with high organic content and insufficient inoculum (Xu et al., 2014, 2015). However, methane concentrations in both C1 and C2 had rapidly increased to over 60% since the aerobic pretreatment was halted. It is noted that only seven and three days were required for the C1 and C2 columns to increase methane concentrations over 60%, respectively. The methane concentrations of C1 and C2 remained relatively stable at between 60% and 70% during the rest of the experiment. The results indicated that the intermittent aerobic pretreatments of MSW aided to establish stable methane production phases without adding anaerobic inoculum. Similar observations can be found elsewhere (Ariunbaatar et al., 2014; Botheju and Bakke, 2011; Xu et al., 2015).

As shown in Fig. 4(b), methane generation rates in C1 and C2 also rapidly increased with methane concentrations in biogas. The methane concentration and the generation rate in C2 improved more rapidly than those in C1, indicating that C2 required less time to transition from the aerobic conditions to the methane generation phase. The higher aeration frequency in C1 might have reduced a portion of the anaerobic zones within the column. As a result, a longer time was required to transition from the aerobic phase to the methanogenic phase in C1.

It was found that the impacts of leachate recirculation on methane production in the C1 and C2 columns were different. When leachate recirculation was resumed, methane generation rates increased again in C2 and reached the highest value (over

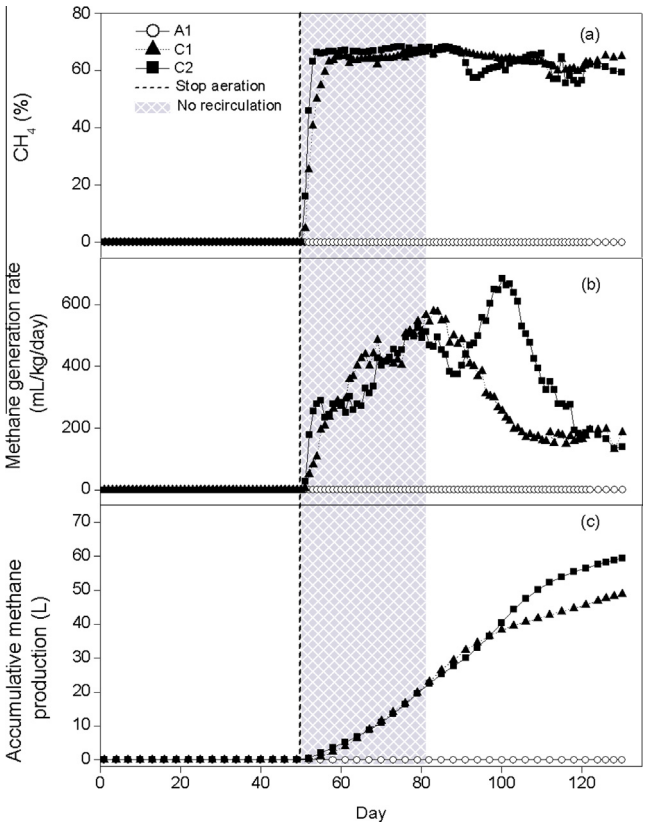


Fig. 4. Methane generations in simulated bioreactors: (a) methane concentrations, (b) daily generation rates and (c) cumulative methane volume.

685 mL/kg/day) at day 100 (Fig. 4(b)); however, a similar trend was observed in C1. With higher methane generation rates, the cumulative methane volume of C2 exceeded that of C1 after day 100 (Fig. 4(c)). These results indicate that an increase in methane generation would be highly dictated by leachate recirculation, as reported in previous studies (Bilgili et al., 2007; Jiang et al., 2007; Sinan et al., 2012; Šan and Onay, 2001; Sanphoti et al., 2006). In spite of leachate recirculation, however, huge changes in the methane generation rates in C1 were not observed; approximately 30 mL/kg/day of methane generation rates additionally increased after leachate recirculation was resumed. These differences, to a large extent, could be attributed to the organic matter that remained in the system after aerobic pretreatment. The greater air supply likely caused degradation of more organic matter in C1 during aerobic pretreatment. As a result, the organic matter that remained in C1 might not be sufficient to enhance methane generation (Charles et al., 2009; Gerassimidou et al., 2013; Zitomer and Shrout, 1998). At the end of the experiment, a total of 49 L and 59 L of methane were collected from C1 and C2, respectively (Fig. 4(c)).

The comparison of methane production between pH-based controlled aeration and oxygen consumption-based control is shown in Table 3. In the study by Xu et al. (2015), aeration was stopped when the leachate pH reached 7.0. However, 60 days (air supplied: 3044 L/kg $_{VS}$) and 75 days (air supplied: 4870 L/kg $_{VS}$) were required to reach pH 7 in both columns with an aeration frequency of 4 times/day and 2 times/day, respectively. With the pH-based aeration control method, the leachate collected from the bottom of the reactors might not properly represent the status of MSW decomposition. Thus it could lead to improperly estimating the aeration period, resulting in losing organic matter for methane recovery. Unlike the previous attempts, in this research, aeration frequencies were adjusted by oxygen consumption rates. Although the initial pH of the leachate was around 5 during 50 days of aerobic pretreatment, methane was rapidly generated in both reactors C1 and C2, as shown in Fig. 4. As seen in Table 3, C2 had the highest cumulative methane generation of 75 L/kg $_{VS}$ among the experimental bioreactor columns for 80 days. As exhibited in the results, a gradual reduction of aeration frequency during aerobic pretreatment of MSW showed great landfill performance with respect to methane recovery from MSW decomposition.

3.4. Implications and limitations

This study focused on conditioning of MSW for enhancing the anaerobic degradation of MSW using gradually reduced aeration. Because MSW is heterogeneous and shows dynamic changes during decomposition, it is difficult to determine aeration parameters for MSW pretreatment. The use of ΔO_2/30 min is a simple approach to control the aeration frequency for the pretreatment. This approach is different from previous quantitative approaches such as operating aeration with a fixed value of oxygen utilization rates. This study's results implied that combining the ΔO_2/30 min patterns with aeration frequency could be used as an indicator of timing for aeration control. However, it should be noted that a two percent reduction of ΔO_2/30 min to adjust the aeration frequency was not tested with other MSW compositions for aerobic pretreatment. Further research should determine a proper standpoint of oxygen demands for various MSW properties. In addition, air distribution in MSW should be considered for controlling aerobic pretreatment, especially at full-scale landfills.

4. Conclusions

Aerobic pretreatment of MSW including high organic content and moisture is an efficient technology to accelerate biogas generation and facilitate waste stabilization in bioreactor landfills. Three landfill-simulated columns were operated for 130 days to investigate the effects of aerobic-anaerobic operation modes on biogas recovery. Two aerobically pretreated bioreactors, including one constant aeration and one gradually reduced aeration frequency, were constructed to investigate the effects of aeration pretreatment modes on methane generation, compared to an anaerobic bioreactor. The oxygen consumption rate after 30 min (ΔO_2/30 min) was used as an indicator for adjusting aeration frequency.

Table 3
Comparison of aeration pretreatment and methane generation.

Aeration control	Aeration pretreatment				pH of leachate at the beginning of anaerobic phase	Accumulative CH_4 generation in anaerobic phase, 80 days (L/kg $_{VS}$)	Reference
	Aeration frequency (times/day)	Flow rate (ml/min-kg $_{VS}$)	Duration (d)	Air volume (L/kg $_{VS}$)			
pH	2	170; 2 h/time	75	3044	7.02	22	Xu et al. (2015)
	4		60	4870	6.95	50	
Oxygen consumption	4	189; 2 h/time	50	4546	5.28	62	This study
	4–2		50	2887	5.02	75	

The results showed that aerobic pretreatment could effectively reduce the impacts of the initial acidogenic phase and adjustment of aeration rates could enhance methane generation. At the end of operation, methane volumes recovered from the C1 and C2 were approximately 62 L/kg $_{VS}$ (44 L/kg$_{dry\ mass}$) and 75 L/kg $_{VS}$ (53 L/kg$_{dry\ mass}$), respectively, while methane produced in the anaerobic control column (A1) was almost negligible. Leachate COD and VFA concentrations were greatly decreased in the aerobically treated bioreactors; an approximately 80% reduction of those constituents was observed compared to the anaerobic bioreactor. In addition, the results indicated that the gradually reduced aeration associated with the oxygen consumption-based approach could be used as an indicator of timing for aeration control. Unlike previous leachate pH-based controlled aeration, the oxygen consumption-based approach reduced the loss of organic matter that might be caused by excess aeration. According to the research, aerobic pretreatment with gradual reduction of aeration rates could not only improve methane recovery from waste decomposition, but also enhance leachate COD and VFA removal.

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